

CAP 1. INTRODUÇÃO ÀS REDES DE COMPUTADORES (AULA 3)

INE5422 REDES DE COMPUTADORES II

PROF. ROBERTO WILLRICH (INE/UFSC)

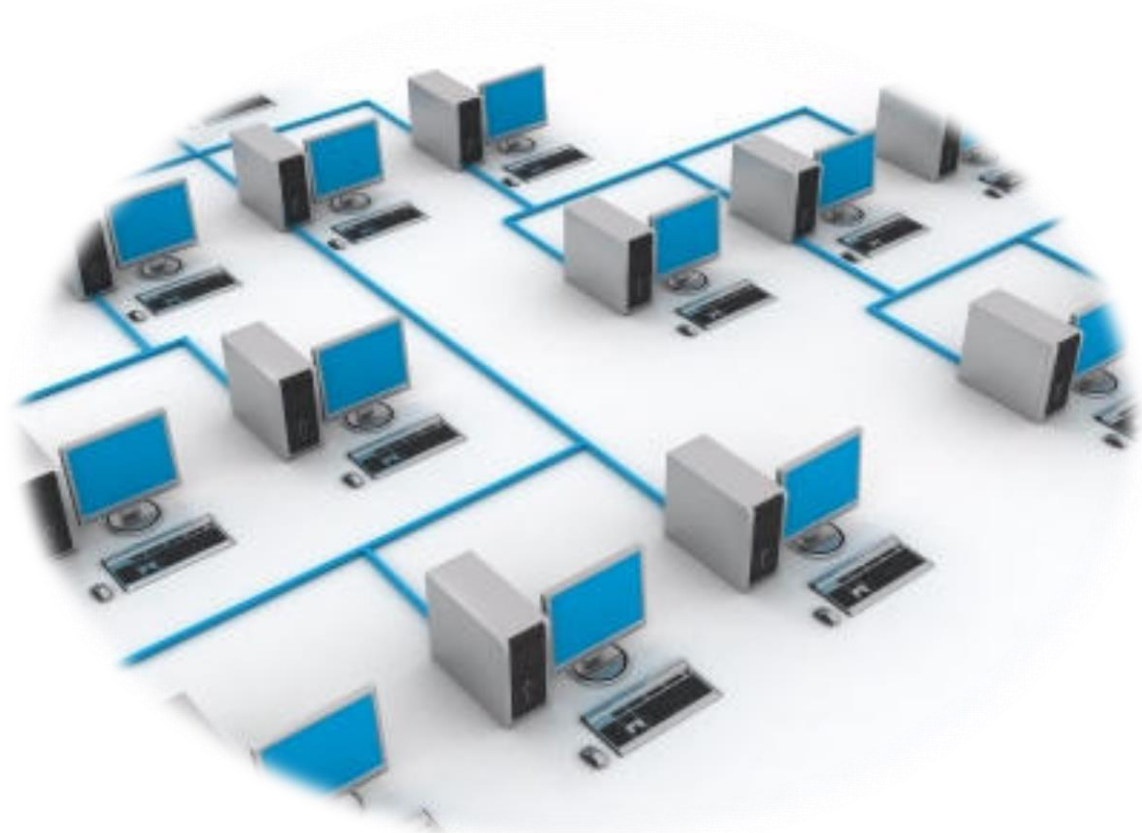
ROBERTO.WILLRICH@UFSC.BR

[HTTPS://MOODLE.UFSC.BR](https://moodle.ufsc.br)

Nesta aula veremos...

O que é Internet

- História da Internet



História da Internet - 1957



- Em 1957 a União Soviética lançou ao espaço o primeiro satélite artificial da Terra, o Sputnik.
- O governo norte-americano então criou o Advanced Research Projects Agency (ARPA), uma divisão do Departamento de Defesa para tentar voltar a liderar a corrida espacial.



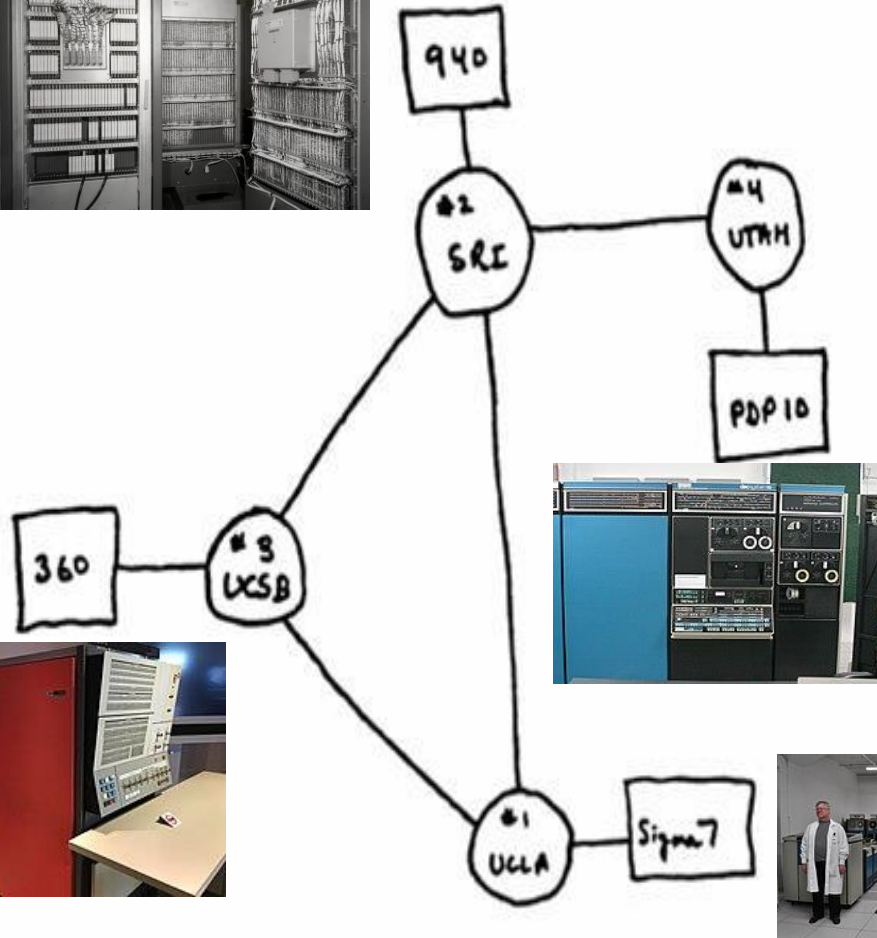
História da Internet - 1962

Foi dado início ao desenvolvimento de uma rede distribuída

- Deveria ser extremamente confiável: que não dependia de um computador central
- Devido a iminência de uma guerra nuclear e o fato de os sistemas de defesa serem extremamente dependente dos computadores



História da Internet - 1968



Sistema de comunicação distribuída entra no ar tendo, no final de 1968:

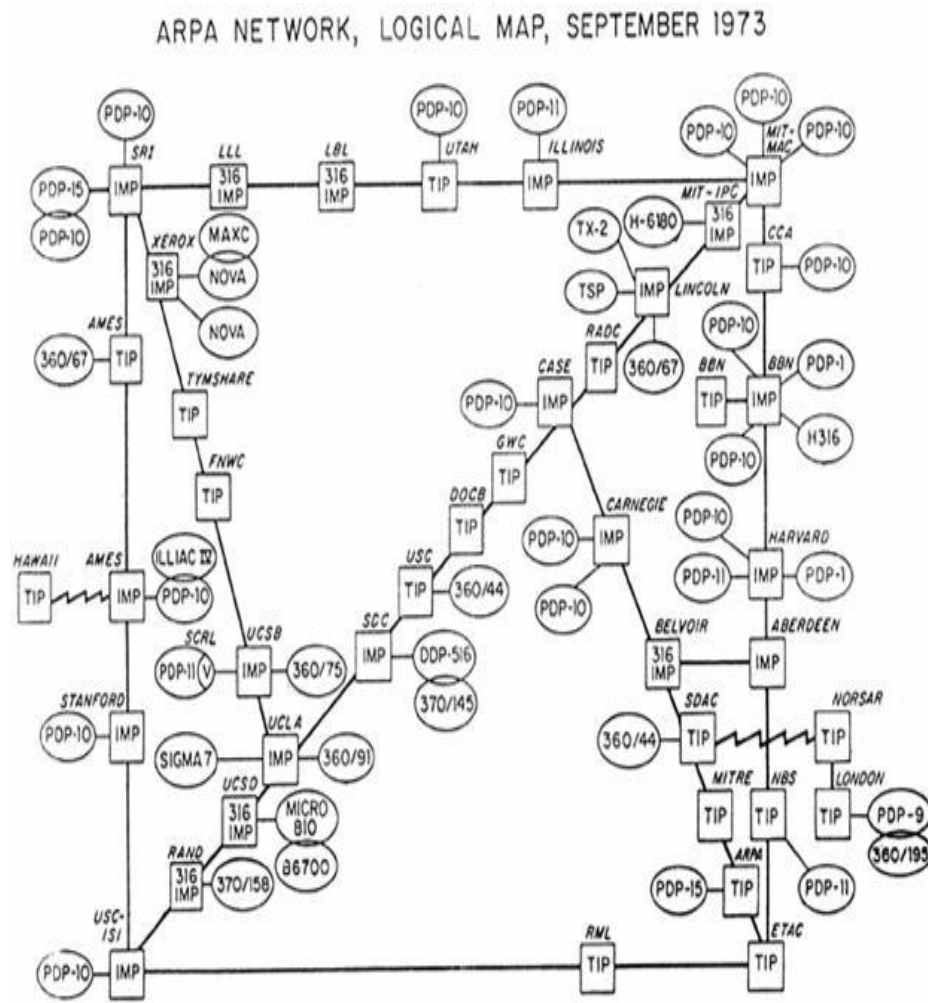
- Quatro computadores conectados
- Velocidade máxima de transferência era de 50 Kbps

Esta criada a ARPANET!



Computadores Conectados: 4

História da Internet - 1973

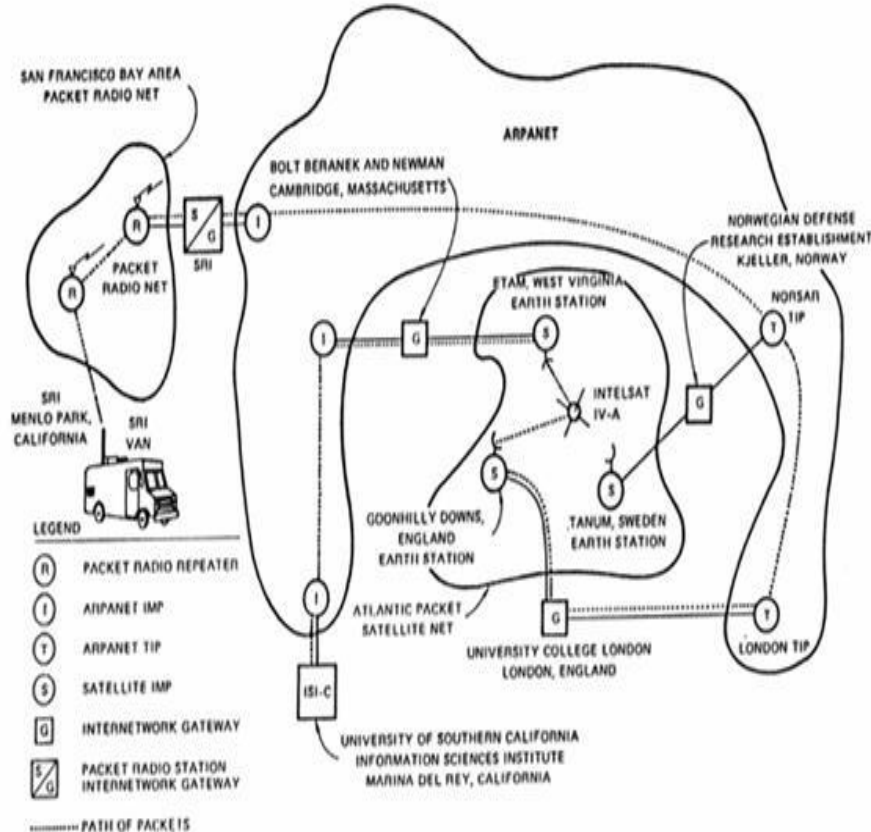


Fatos

- Desenvolvido o primeiro programa de e-mail
- Primeiro computador internacional conectado (Inglaterra) via rádio
- TCP/IP começa a ser desenvolvido
 - Necessidade de padronização de comunicação entre outras redes e a ARPANET

Computadores Conectados: 23

História da Internet - 1977

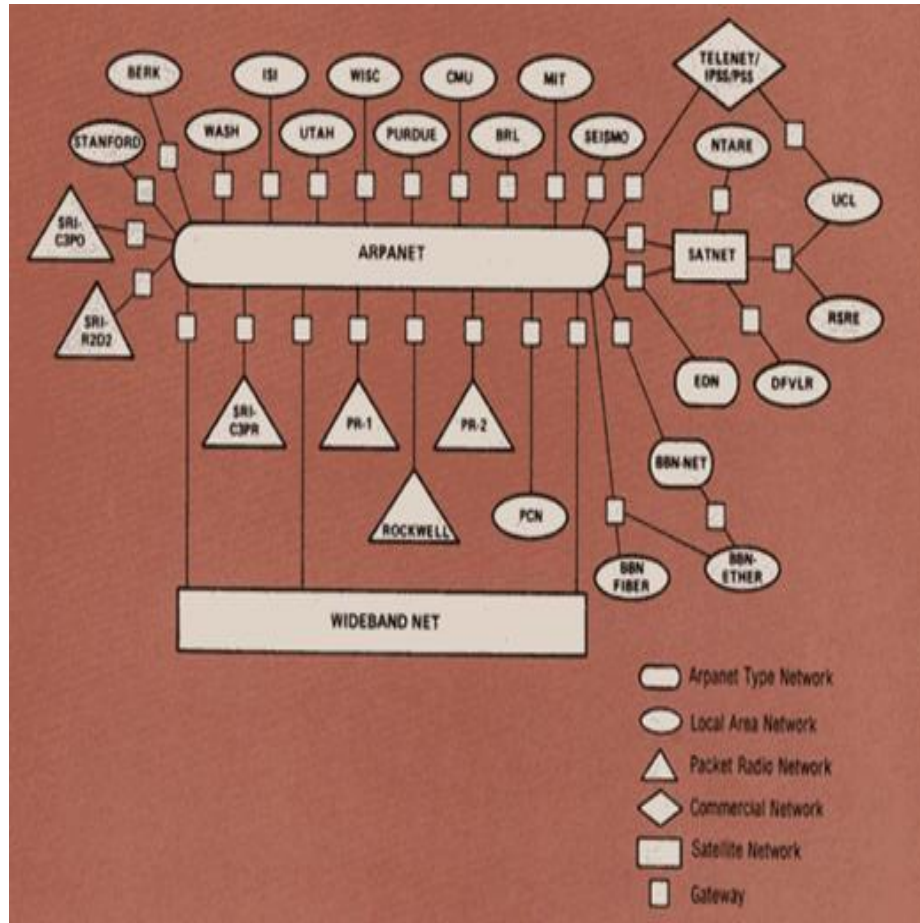


Fatos

- Protocolo TCP/IP desenvolvido e incorporado no sistema Unix.
- Aumento da velocidade da rede com novos cabos de comunicação.
- Conexão via satélite na rede com o SATNET.
- **Integração de outras redes à ARPANET, início da internetting**
- Primeira conexão do Brasil com a rede ARPANET via ligação telefônica.

Computadores Conectados: 110

História da Internet - 1983



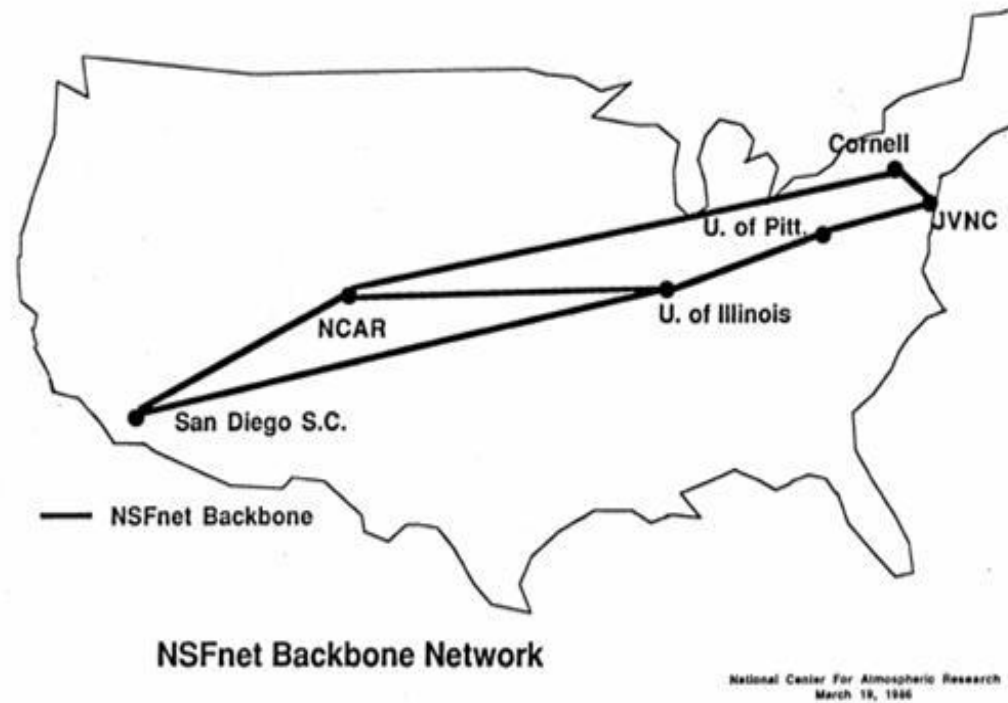
TCP/IP se torna padrão

Criado o DNS (Domain Name Service) pela universidade de Wisconsin

- permitindo que cada computador tenha um nome ao invés de um número

Computadores Conectados: 405

História da Internet - 1984



Fatos

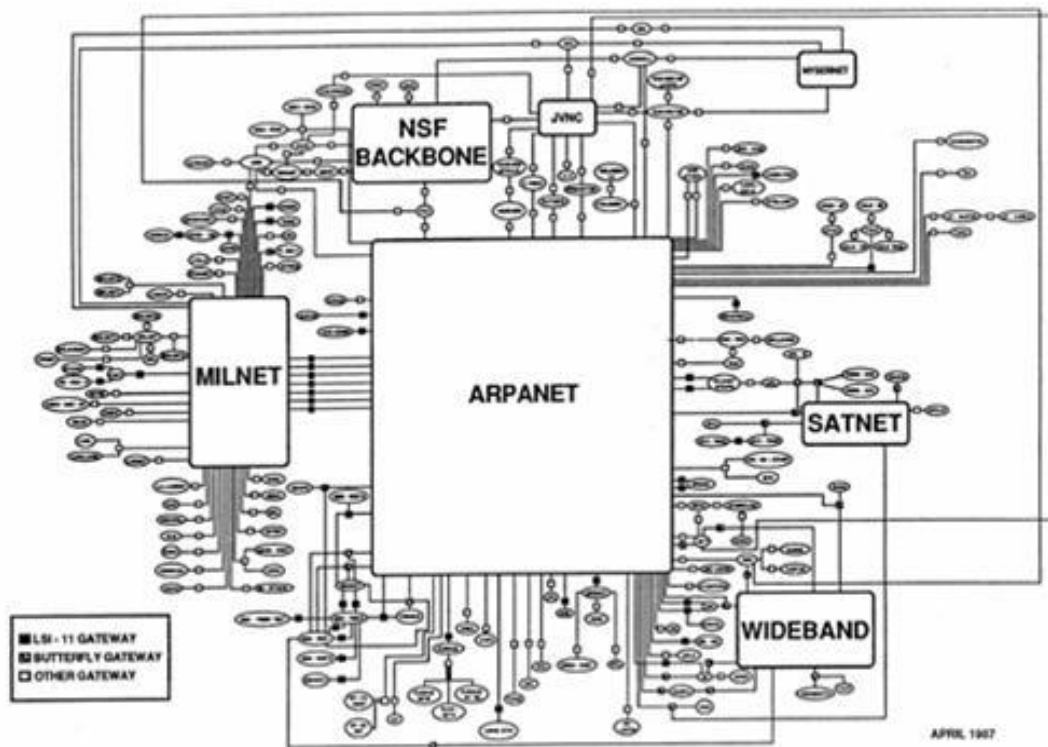
- ARPANET dividida em ARPANET e MILNET
- Velocidade de conexão sobe para 1.5Mbps
- Introdução de sufixos nos nomes (.com,.org,.gov,.mil,.edu)

Computadores Conectados: 1.000

História da Internet - 1988

Fatos

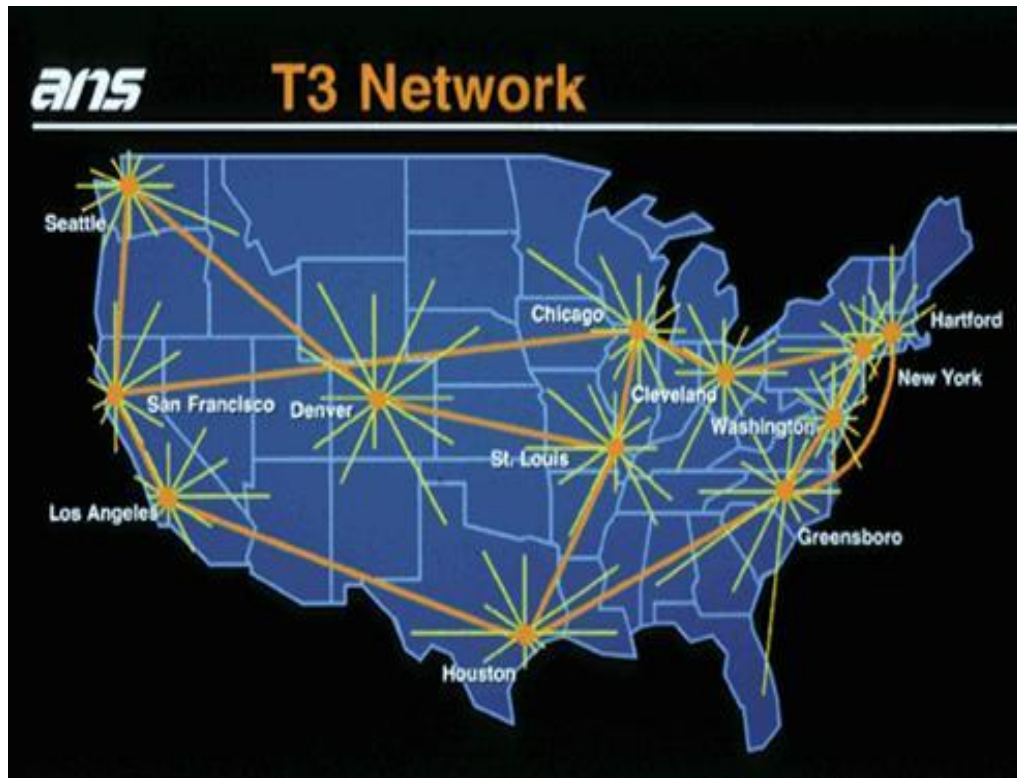
- Devido ao grande crescimento do número de computadores conectados (mais de 56.000) surge a necessidade de aumentar ainda mais a velocidade da rede.
- Conexão de outros países como Canadá, Finlândia, França, Dinamarca, etc...
- Primeiro Vírus na rede (1986).
- Primeira reunião para o estabelecimento de uma rede para fins acadêmicos no Brasil.



BBN Communications Corporation

Computadores Conectados: 56.000

História da Internet - 1991



Fatos

- DARPA fecha definitivamente a rede ARPANET em 1990
- Mais países são conectados (Brasil, Argentina, Chile, Bélgica, Índia, dentre outros)
- É criado o conceito de HTML: Início da World Wide Web - WWW

Computadores Conectados: 600.000

Internet: Hoje

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ESSENTIAL DIGITAL HEADLINES

OVERVIEW OF THE ADOPTION AND USE OF CONNECTED DEVICES AND SERVICES



TOTAL
POPULATION



we
are
social

8.20
BILLION

URBANISATION

58.1%

UNIQUE MOBILE
PHONE SUBSCRIBERS



Meltwater

5.78
BILLION

vs. POPULATION

70.5%

INDIVIDUALS USING
THE INTERNET



KEPIOS

5.56
BILLION

vs. POPULATION

67.9%

SOCIAL MEDIA
USER IDENTITIES

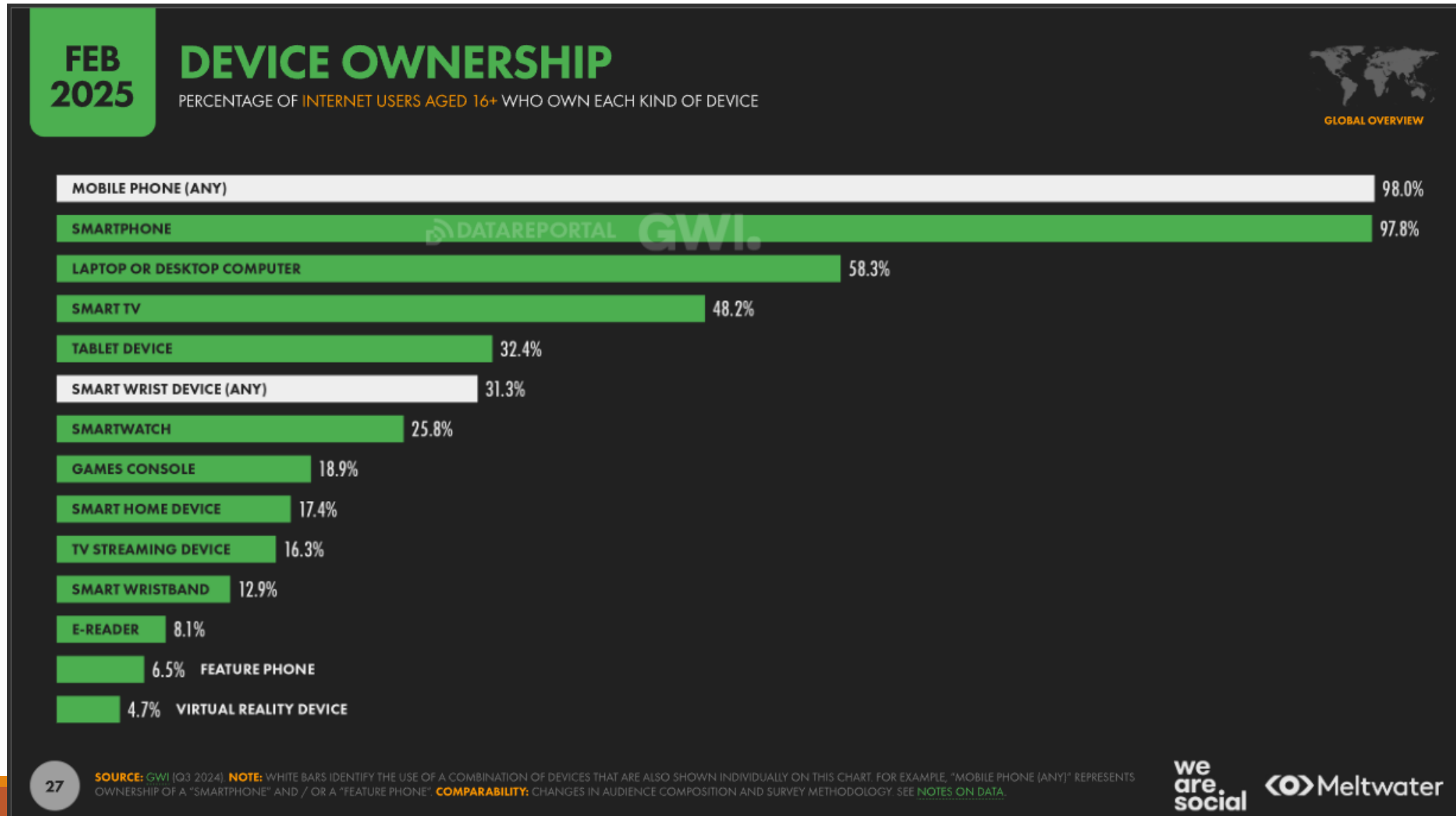


5.24
BILLION

vs. POPULATION

63.9%

Internet: Hoje

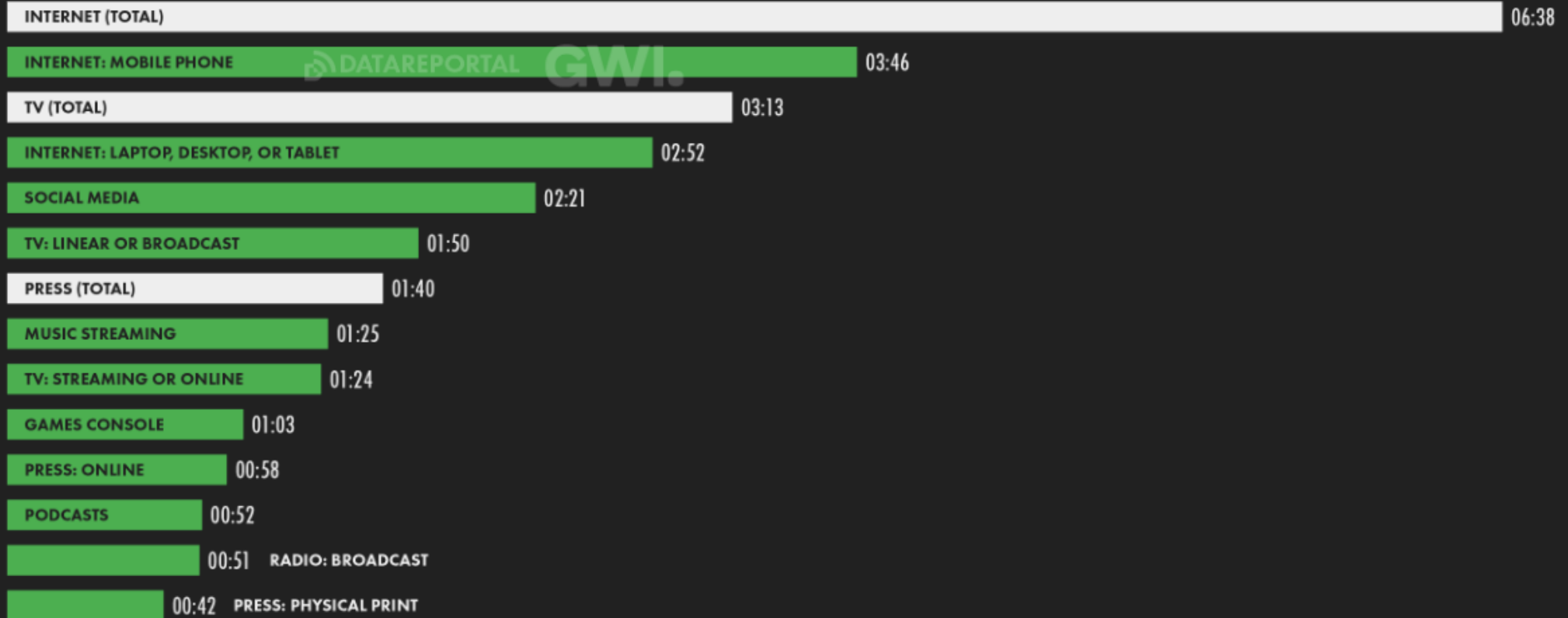


Internet: Hoje

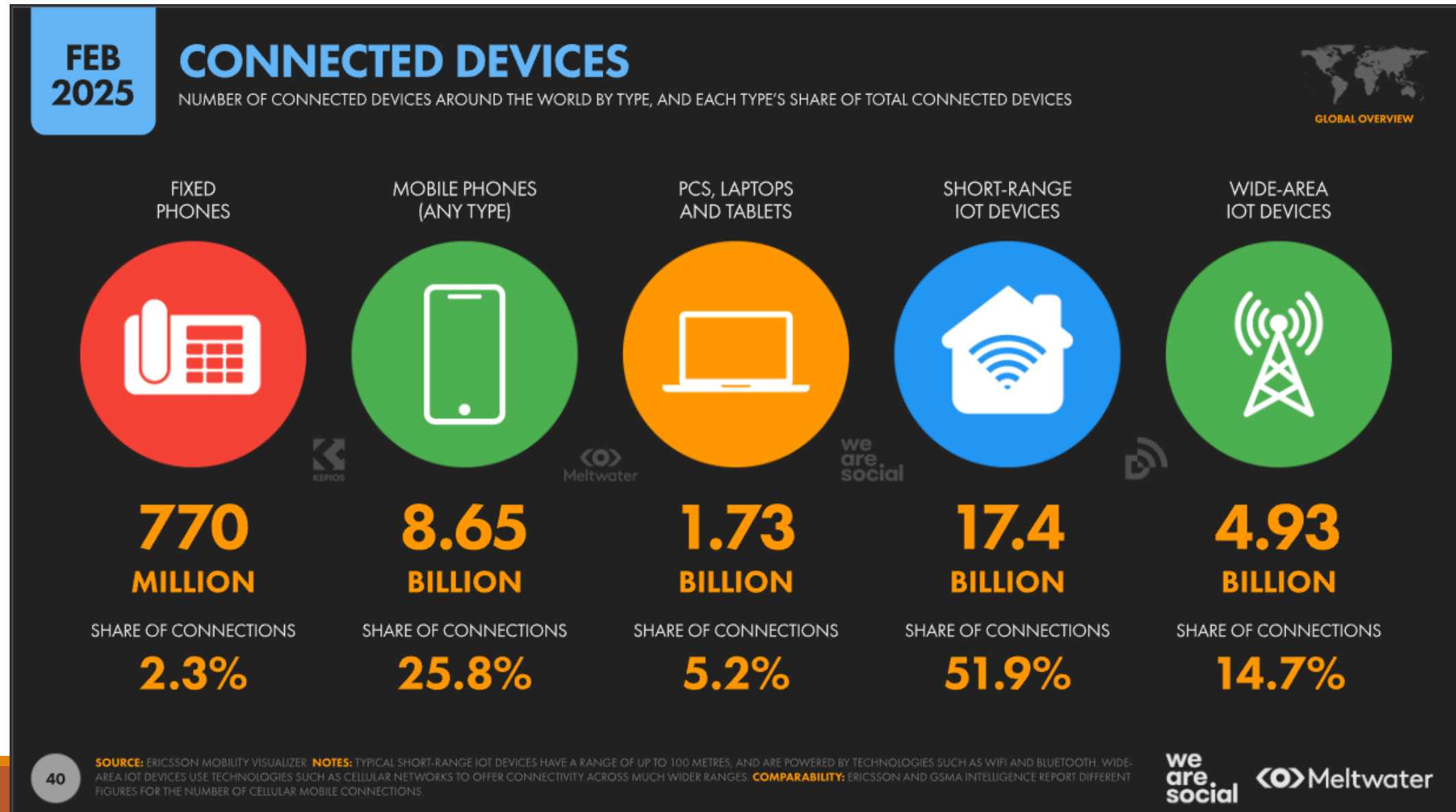
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DAILY TIME SPENT WITH MEDIA

THE AVERAGE AMOUNT OF TIME (IN HOURS AND MINUTES) THAT **INTERNET USERS AGED 16+** SPEND WITH EACH MEDIUM OR DEVICE EACH DAY



Internet: Hoje



Internet: Hoje

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OVERVIEW OF INTERNET USE

ESSENTIAL INDICATORS OF INTERNET ADOPTION AND USE



INDIVIDUALS
USING THE
INTERNET



5.56
BILLION



INDIVIDUALS USING THE
INTERNET AS A PERCENTAGE
OF TOTAL POPULATION



67.9%
YOY: +1.6% (+109 BPS)



YEAR-ON-YEAR CHANGE IN
THE NUMBER OF INDIVIDUALS
USING THE INTERNET



+2.5%
+136 MILLION



PERCENTAGE OF THE
TOTAL FEMALE POPULATION
THAT USES THE INTERNET



65.7%
YOY: +3.7% (+232 BPS)



PERCENTAGE OF THE
TOTAL MALE POPULATION
THAT USES THE INTERNET



70.0%
YOY: +2.7% (+181 BPS)

AVERAGE DAILY TIME
SPENT USING THE INTERNET
BY EACH INTERNET USER



6H 38M
YOY: -0.4% (-2 MINS)



PERCENTAGE OF USERS
ACCESSING THE INTERNET
VIA MOBILE PHONES



96.3%
YOY: -0.2% (-20 BPS)



PERCENTAGE OF USERS
ACCESSING THE INTERNET
VIA LAPTOPS AND DESKTOPS



61.5%
YOY: -0.5% (-30 BPS)



PERCENTAGE OF THE
TOTAL URBAN POPULATION
THAT USES THE INTERNET



82.7%
YOY: +3.6% (+289 BPS)

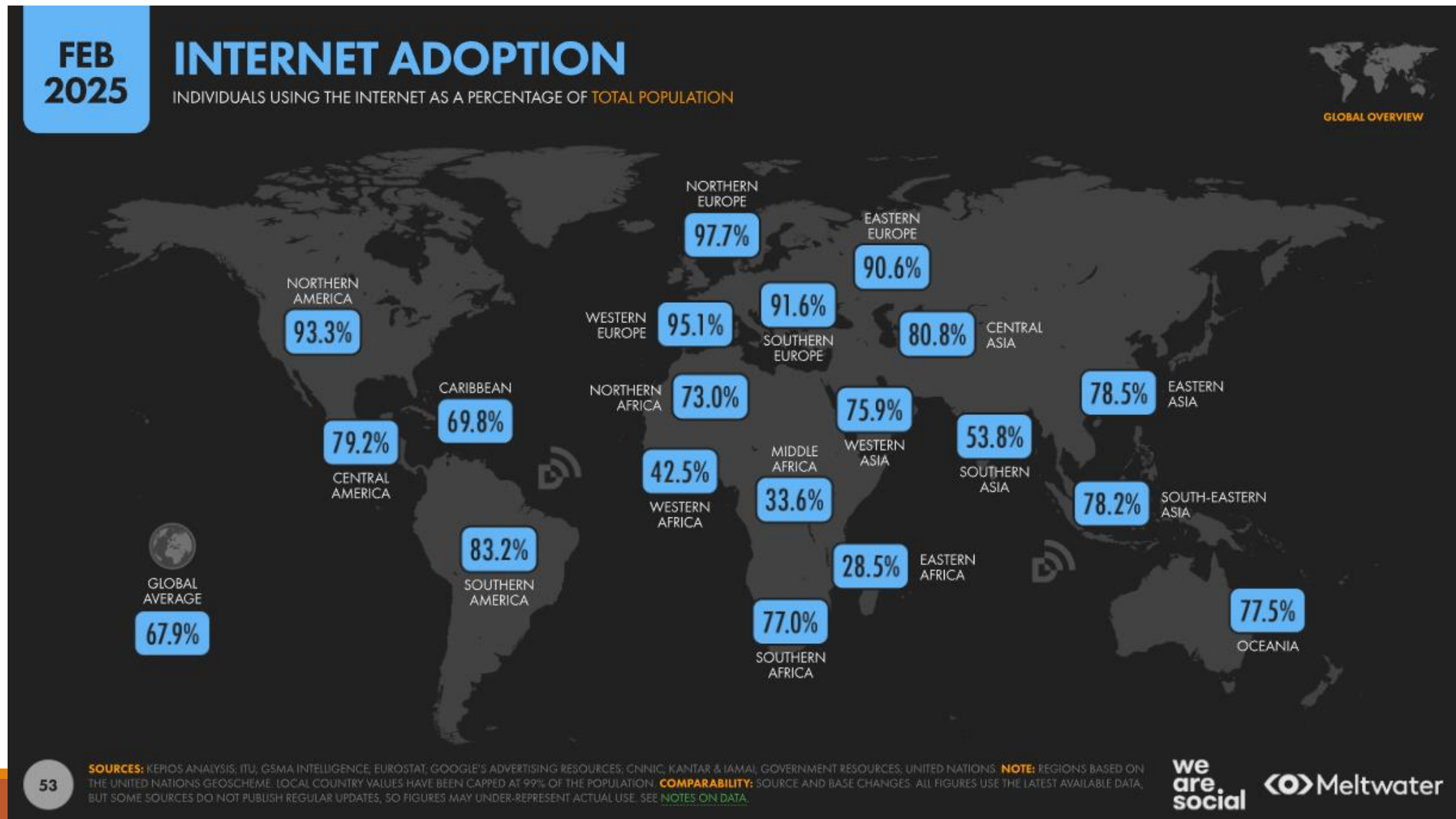


PERCENTAGE OF THE
TOTAL RURAL POPULATION
THAT USES THE INTERNET

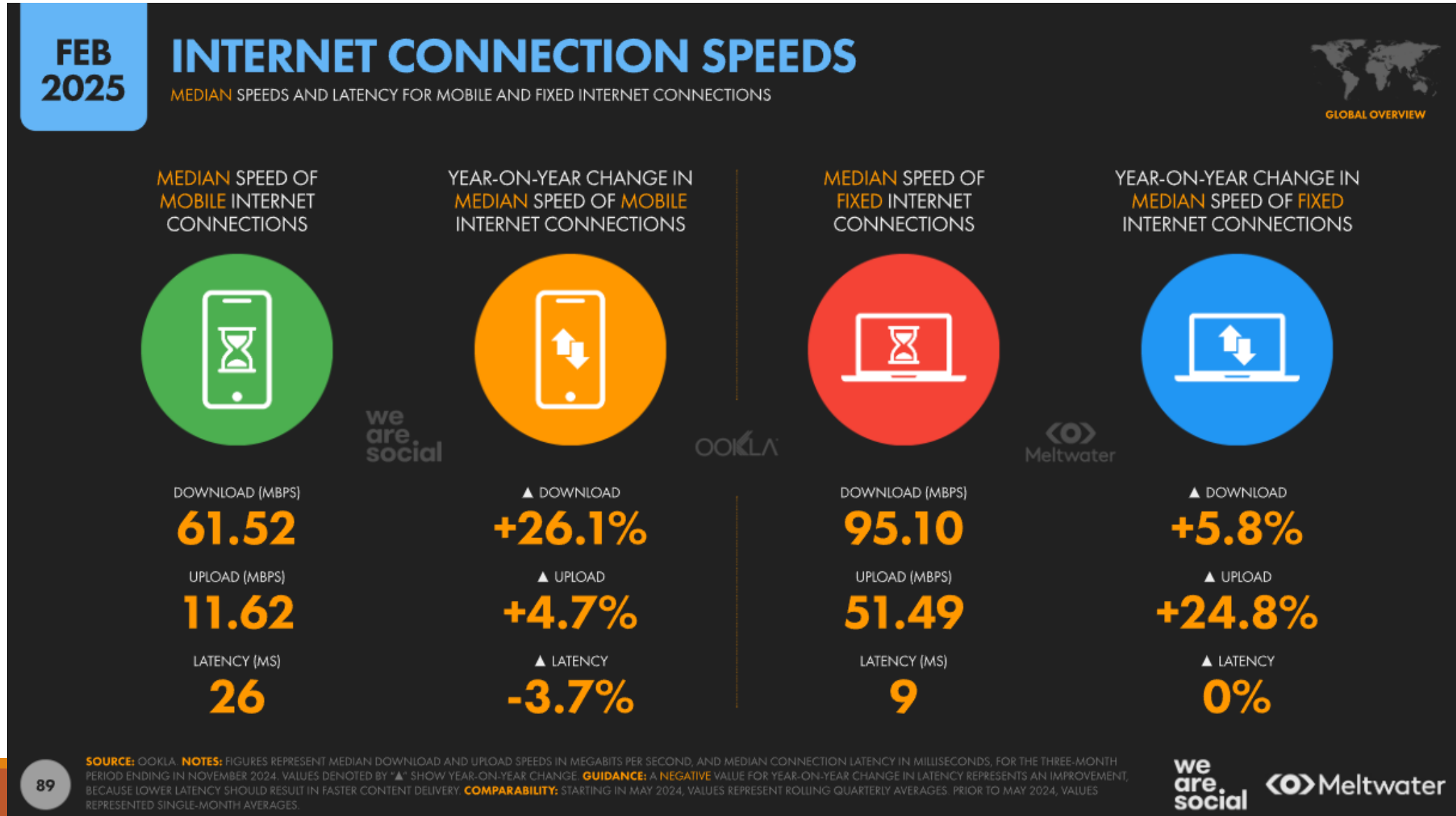


47.4%
YOY: +2.8% (+130 BPS)

Internet: Hoje



Internet: Hoje



Internet: Hoje

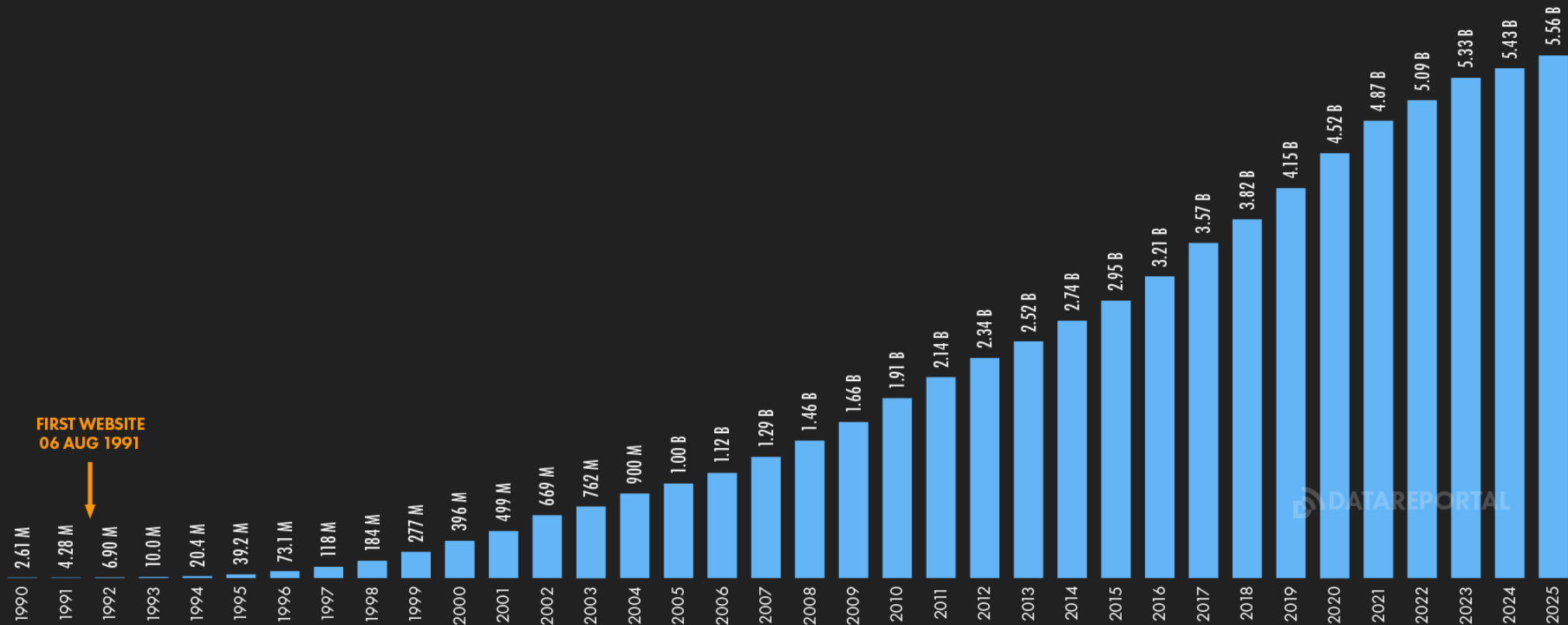
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INTERNET USE TIMELINE

NUMBER OF INDIVIDUALS USING THE INTERNET OVER TIME



GLOBAL OVERVIEW

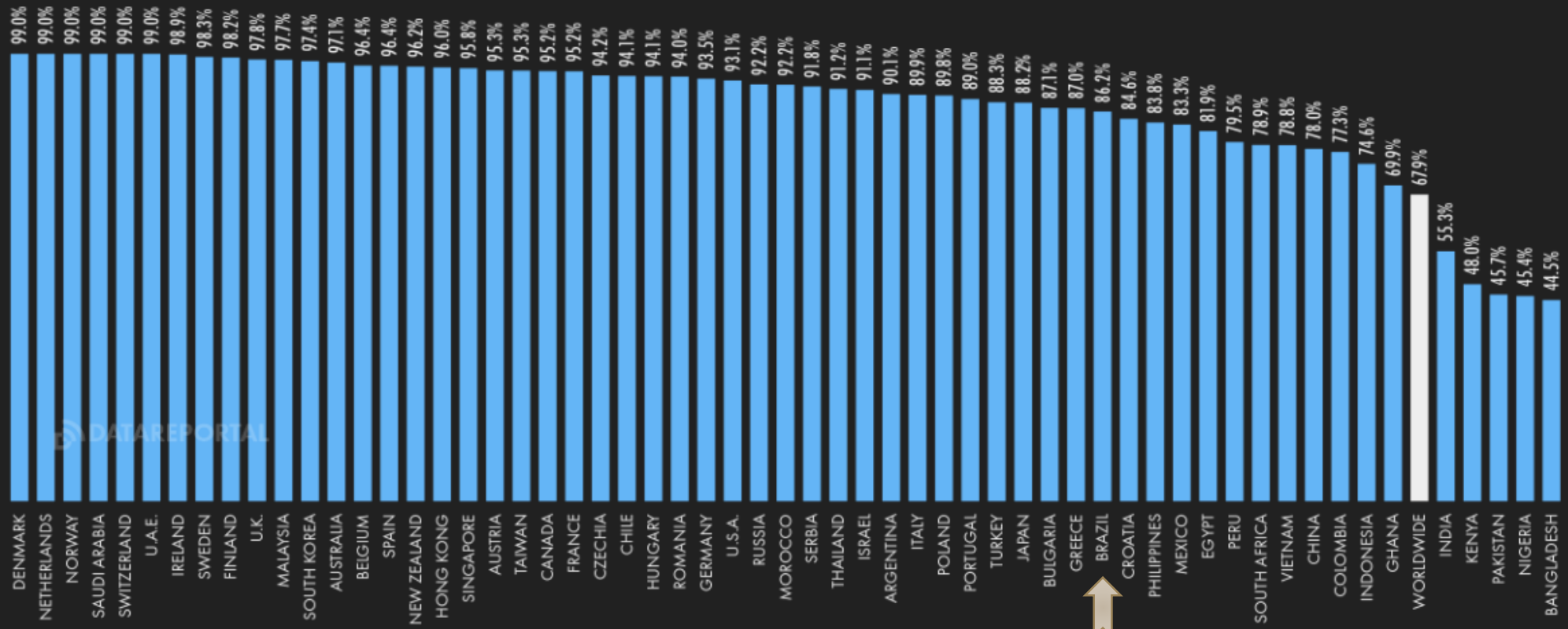


Internet: Hoje

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INTERNET ADOPTION

INDIVIDUALS USING THE INTERNET AS A PERCENTAGE OF TOTAL POPULATION



Internet: Hoje

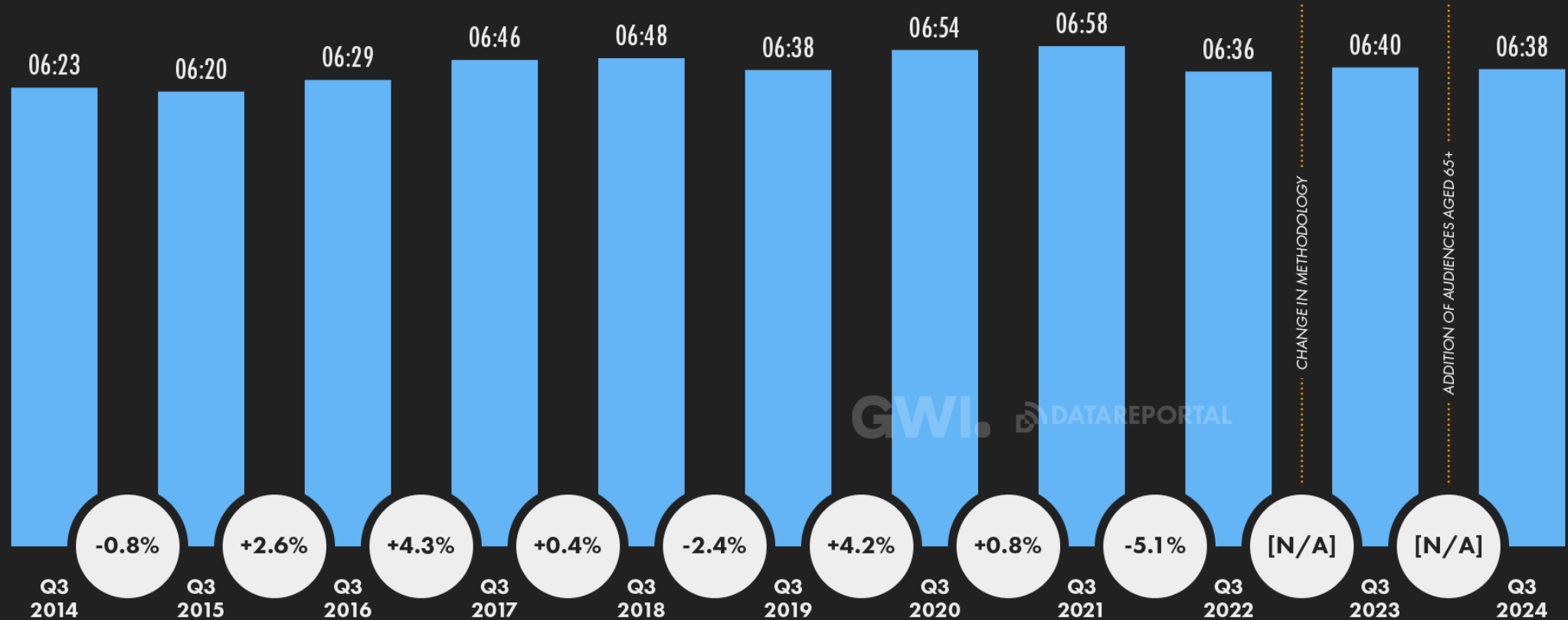
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DAILY TIME SPENT USING THE INTERNET (YOY)

THE AVERAGE AMOUNT OF TIME (IN HOURS AND MINUTES) THAT INTERNET USERS AGED 16+ SPEND USING THE INTERNET EACH DAY



GLOBAL OVERVIEW



Internet: Hoje

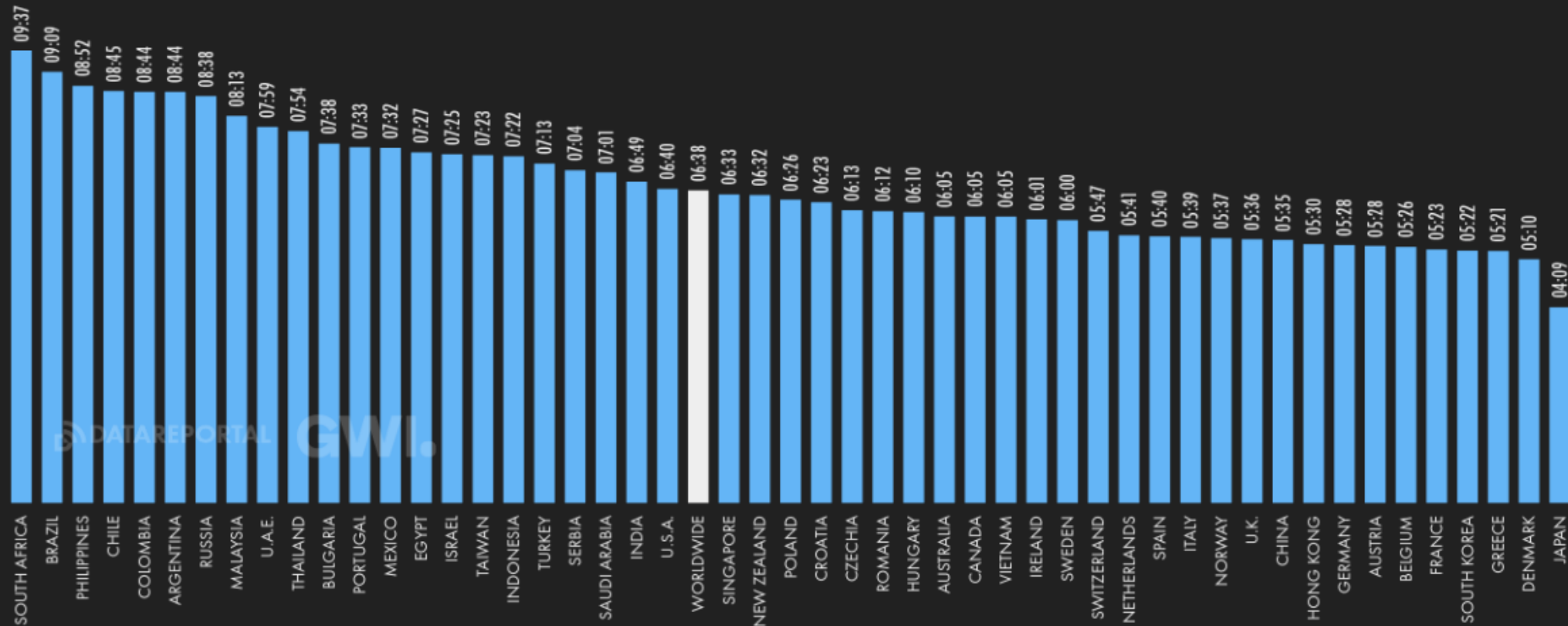
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DAILY TIME SPENT USING THE INTERNET

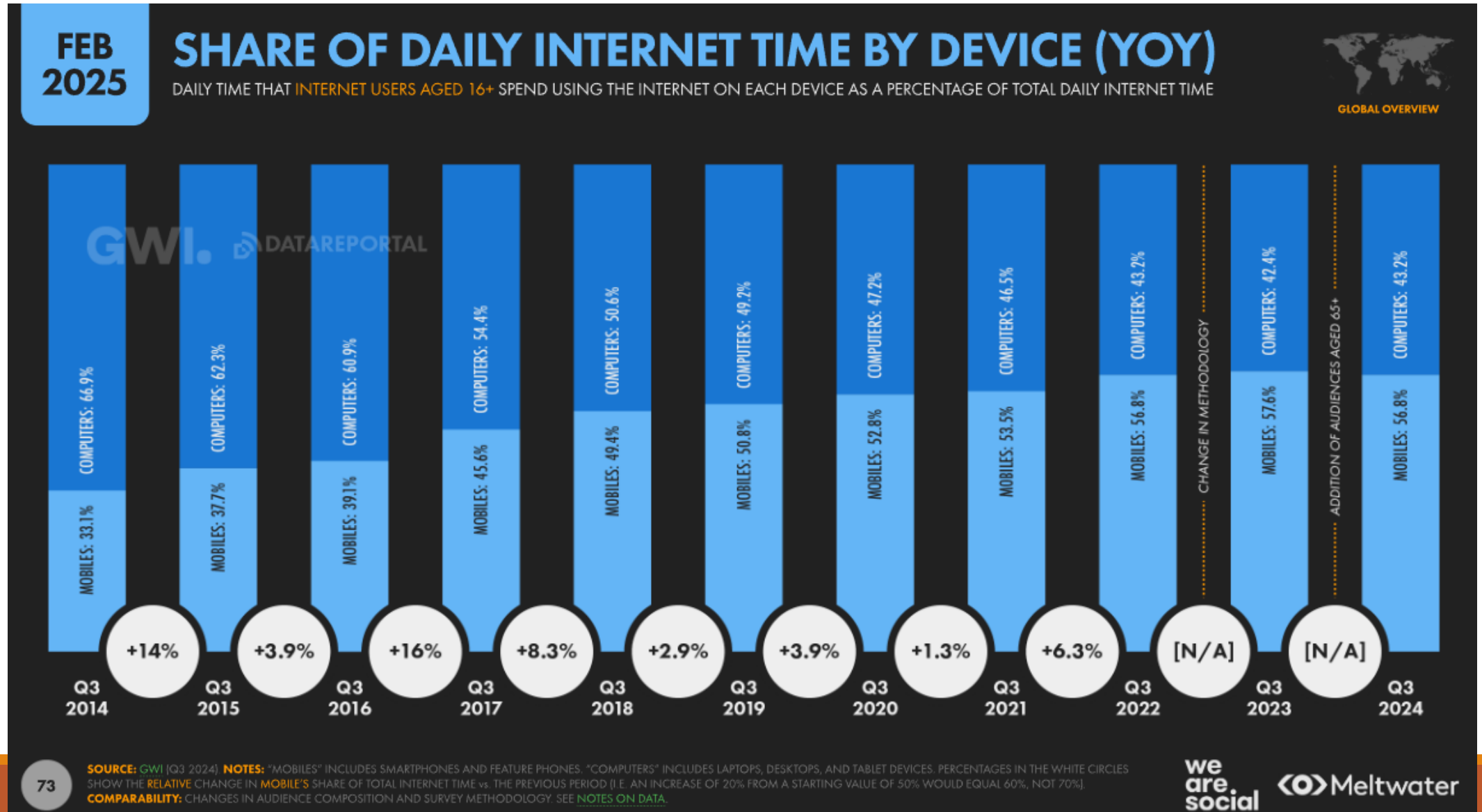
AVERAGE AMOUNT OF TIME (IN HOURS AND MINUTES) THAT INTERNET USERS AGED 16+ SPEND USING THE INTERNET EACH DAY ON ANY DEVICE



GLOBAL OVERVIEW



Internet: Hoje

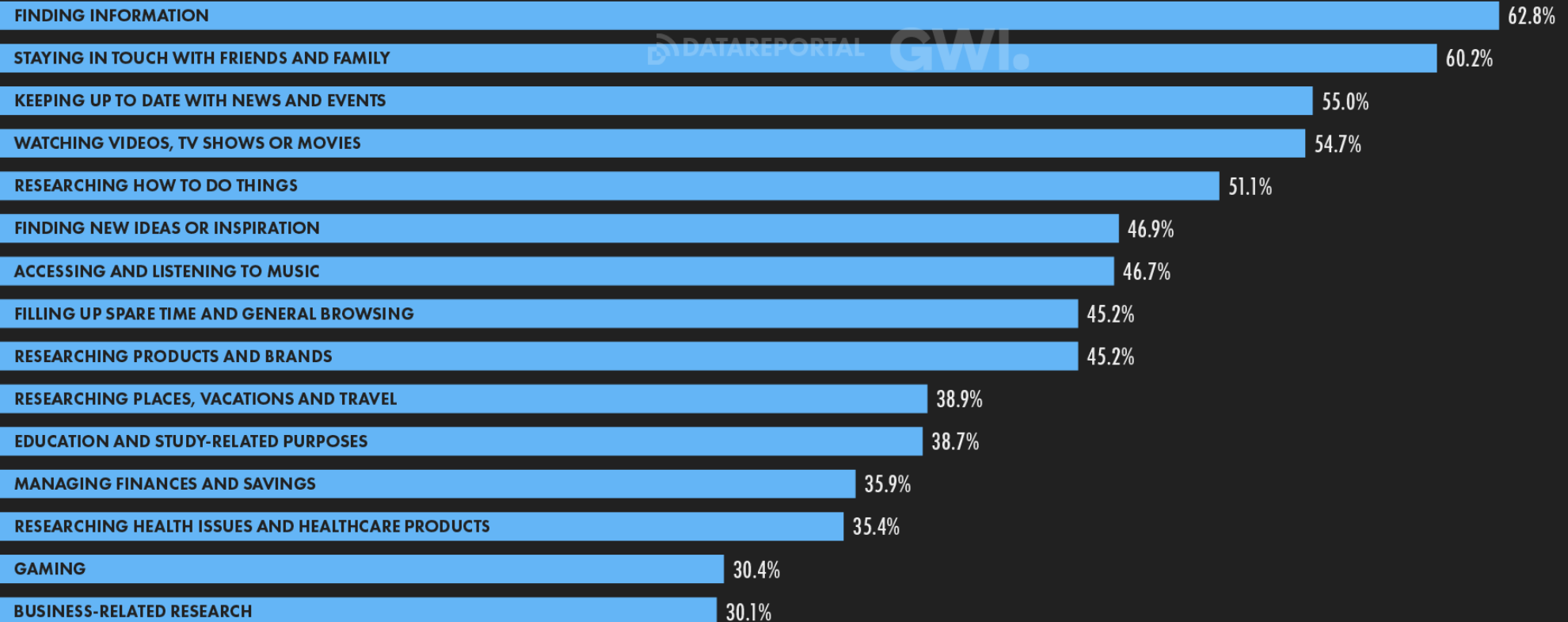


Internet: Hoje

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MAIN REASONS FOR USING THE INTERNET

PRIMARY REASONS WHY INTERNET USERS AGED 16+ USE THE INTERNET



Internet: Hoje

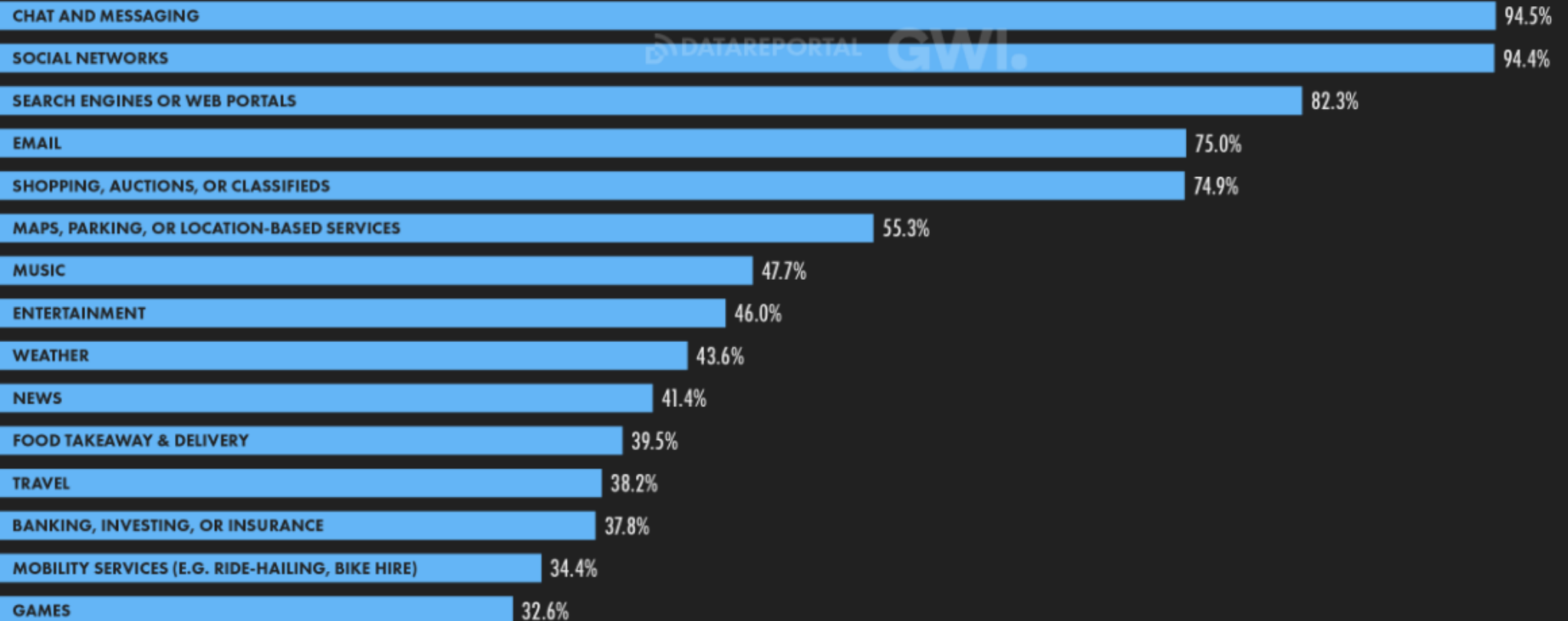
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TOP TYPES OF WEBSITES VISITED AND APPS USED

PERCENTAGE OF INTERNET USERS AGED 16+ WHO HAVE VISITED OR USED EACH KIND OF DIGITAL PROPERTY IN THE PAST MONTH



GLOBAL OVERVIEW



Internet: Hoje

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TOP WEBSITES: SIMILARWEB RANKING

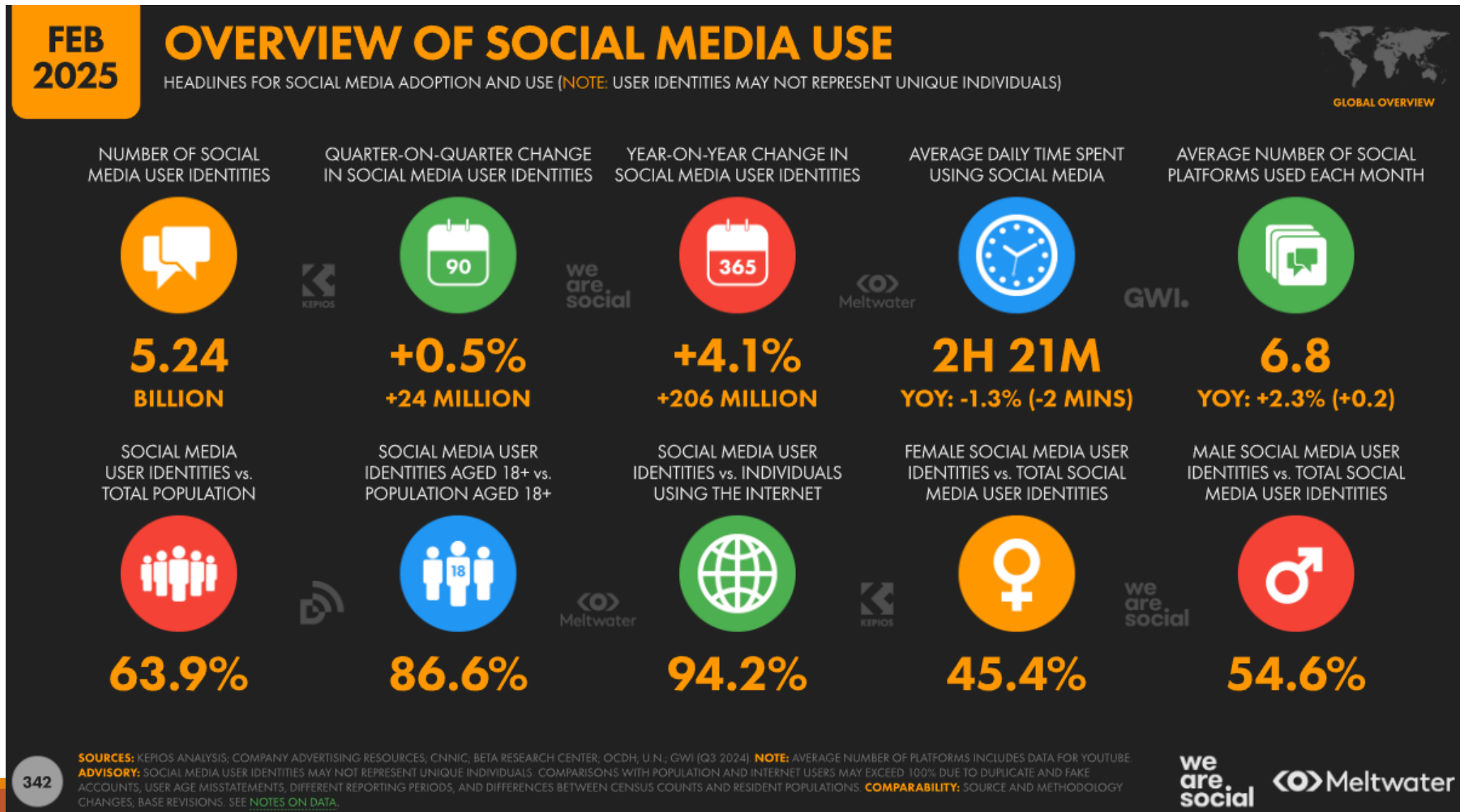
SIMILARWEB'S RANKING OF THE MOST VISITED WEBSITES, BASED ON WEBSITE TRAFFIC BETWEEN 01 SEPTEMBER AND 30 NOVEMBER 2024



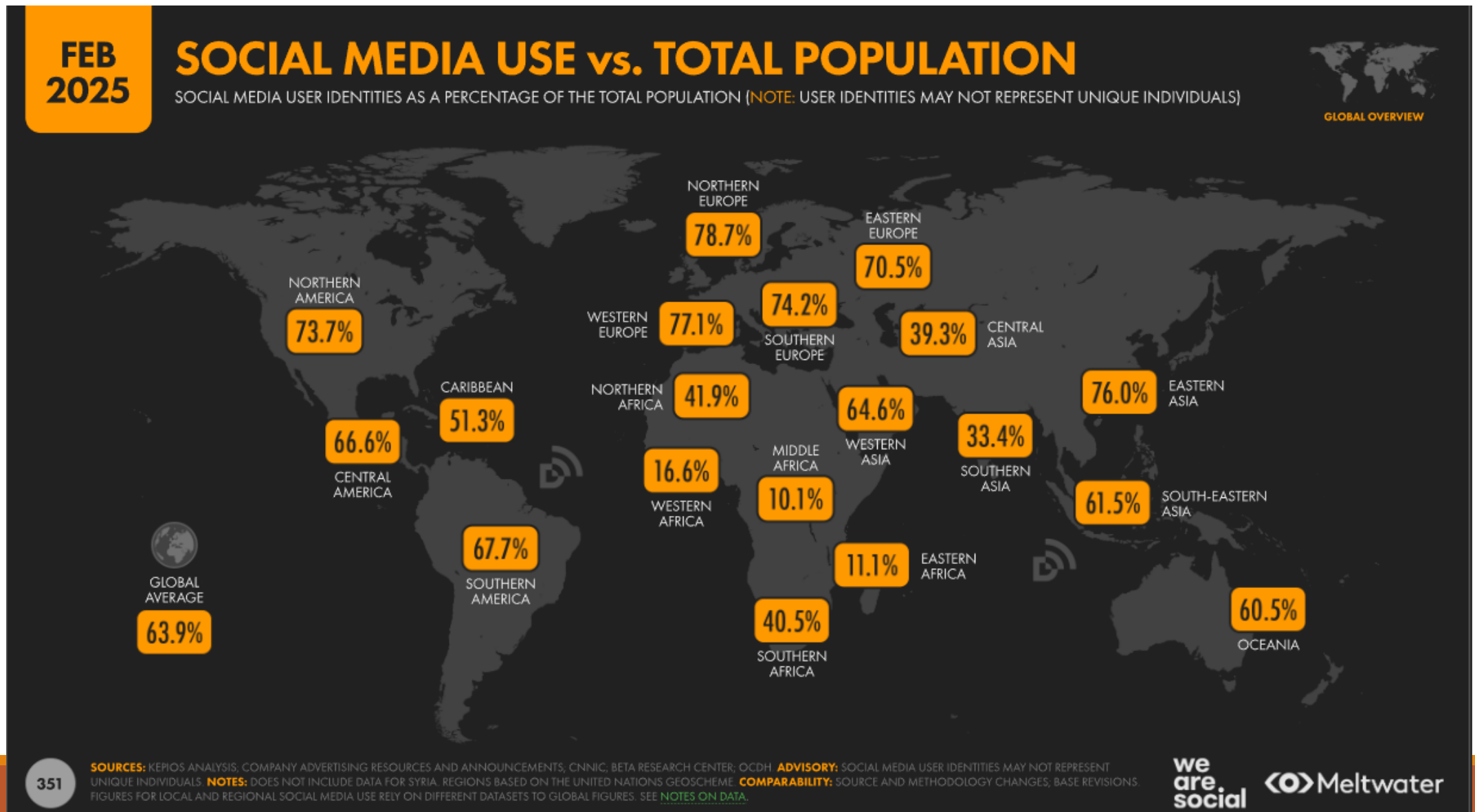
#	WEBSITE	TOTAL VISITS (MONTHLY AVE.)	UNIQUE VISITORS (MONTHLY AVE.)	AVERAGE TIME PER VISIT	AVERAGE PAGES PER VISIT
01	GOOGLE.COM	83.0 B	3.17 B	10M 52S	8.18
02	YOUTUBE.COM	28.5 B	1.73 B	20M 07S	10.90
03	FACEBOOK.COM	12.3 B	1.04 B	10M 53S	11.66
04	INSTAGRAM.COM	5.88 B	931 M	8M 38S	11.60
05	WHATSAPP.COM	4.55 B	596 M	15M 10S	8.27
06	X.COM	4.44 B	448 M	12M 17S	12.20
07	WIKIPEDIA.ORG	3.93 B	908 M	3M 19S	3.11
08	CHATGPT.COM	3.53 B	310 M	6M 12S	3.67
09	REDDIT.COM	3.50 B	642 M	6M 02S	4.60
10	YAHOO.COM	3.45 B	403 M	8M 04S	5.23

#	WEBSITE	TOTAL VISITS (MONTHLY AVE.)	UNIQUE VISITORS (MONTHLY AVE.)	AVERAGE TIME PER VISIT	AVERAGE PAGES PER VISIT
11	YAHOO.CO.JP	3.22 B	142 M	9M 11S	7.49
12	YANDEX.RU	2.80 B	180 M	8M 51S	8.37
13	AMAZON.COM	2.69 B	515 M	6M 16S	9.13
14	BAIDU.COM	2.40 B	138 M	5M 38S	8.36
15	TIKTOK.COM	2.20 B	743 M	3M 49S	4.95
16	NETFLIX.COM	1.83 B	252 M	6M 57S	3.75
17	MICROSOFTONLINE.COM	1.82 B	286 M	2M 17S	2.65
18	BING.COM	1.78 B	218 M	6M 51S	5.13
19	PORNHUB.COM	1.73 B	284 M	8M 00S	7.61
20	LINKEDIN.COM	1.70 B	318 M	8M 17S	7.78

Internet: Hoje



Internet: Hoje

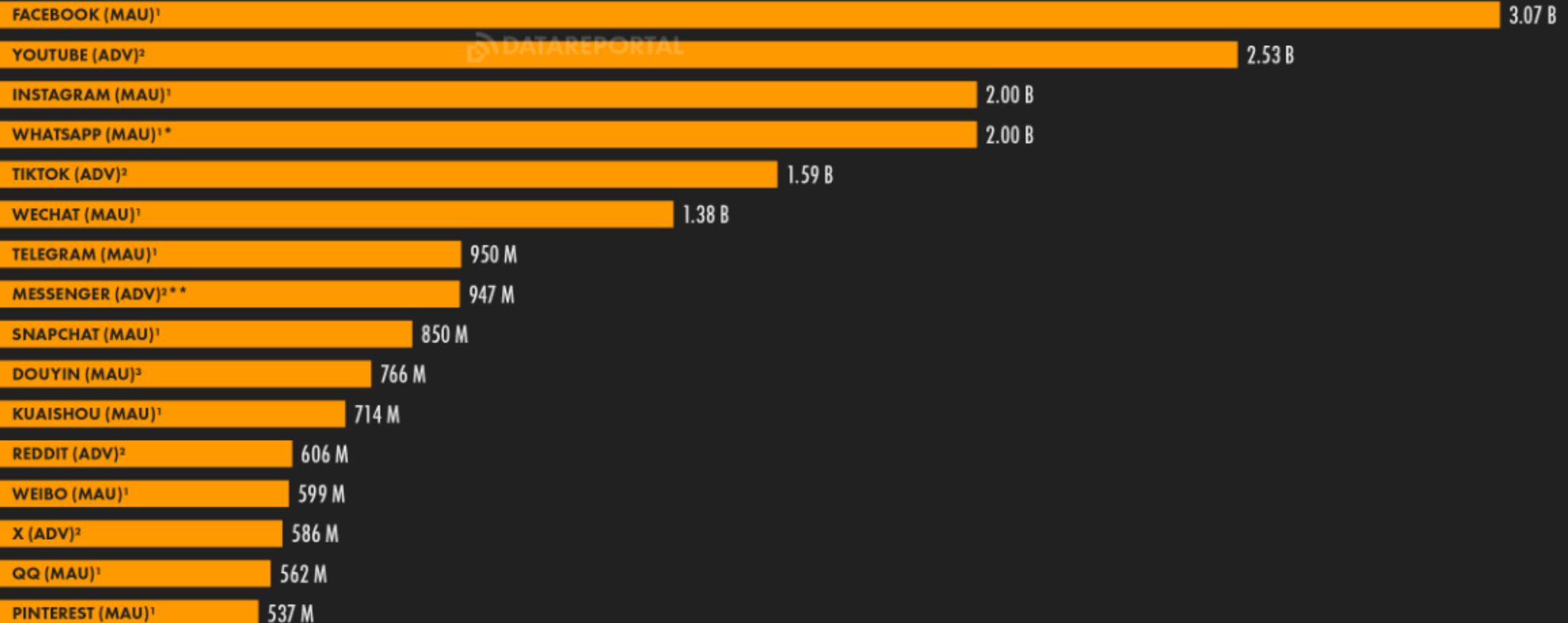


Internet: Hoje

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PERSPECTIVES: SOCIAL MEDIA PLATFORM USE

A MIX OF METRICS ILLUSTRATING USE OF THE WORLD'S TOP SOCIAL MEDIA PLATFORMS (NOTE: VALUES MAY NOT REPRESENT UNIQUE INDIVIDUALS)

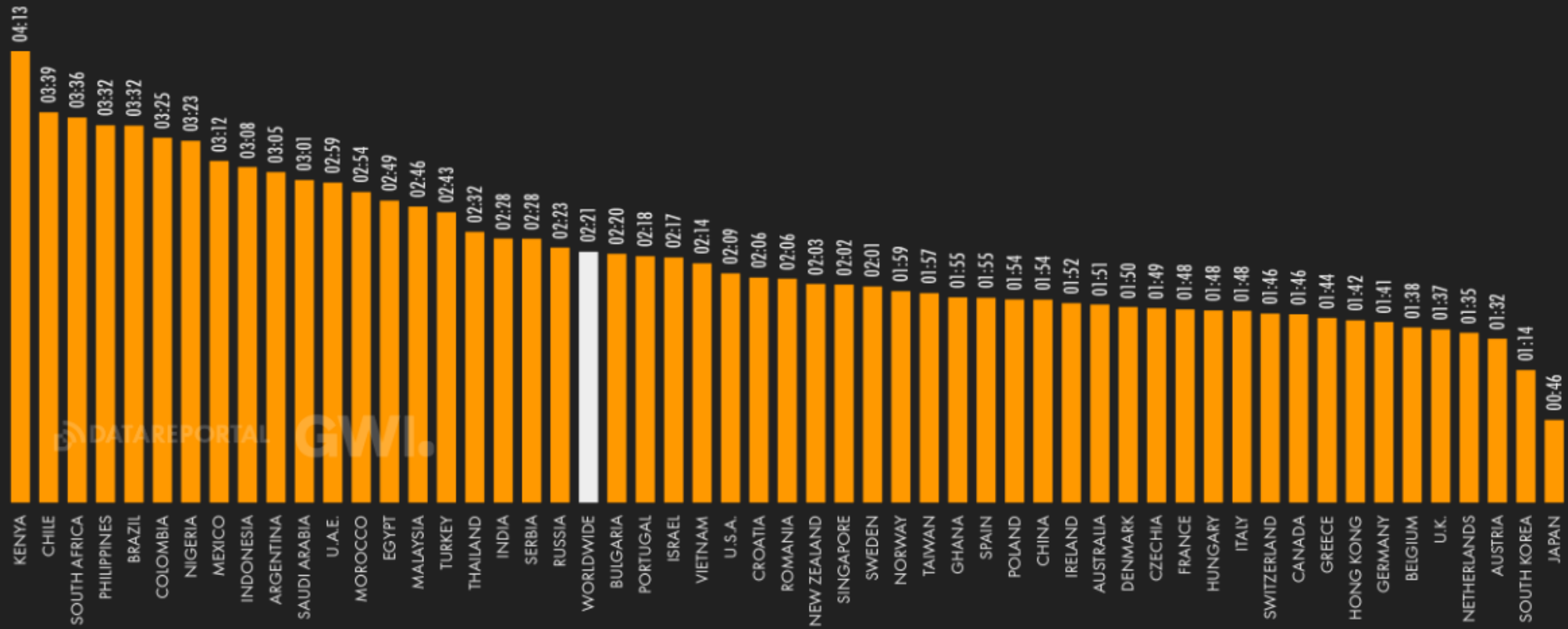


Internet: Hoje

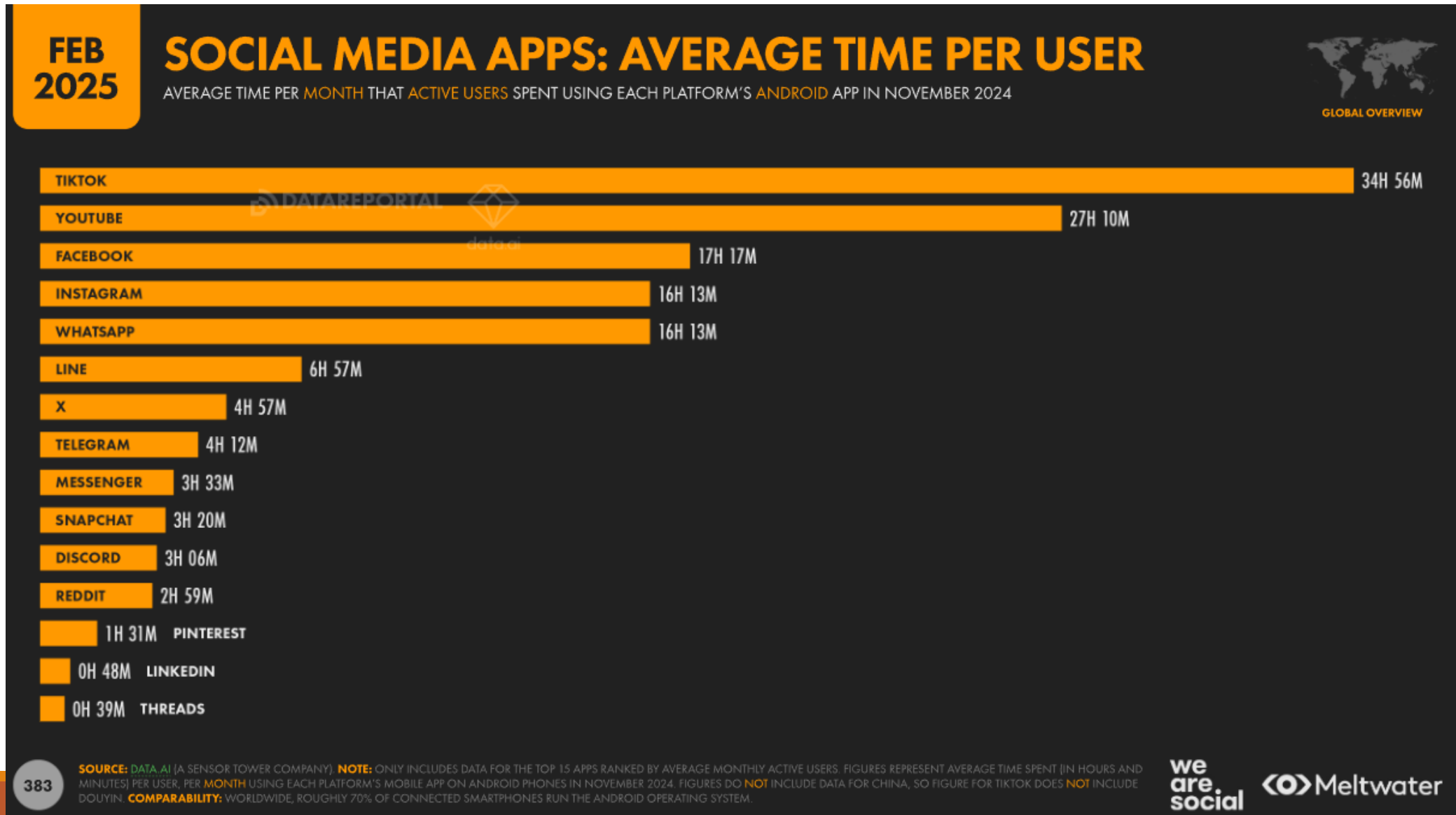
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DAILY TIME SPENT USING SOCIAL MEDIA

AVERAGE AMOUNT OF TIME (IN HOURS AND MINUTES) THAT INTERNET USERS AGED 16+ SPEND USING SOCIAL MEDIA EACH DAY



Internet: Hoje



Internet: Hoje

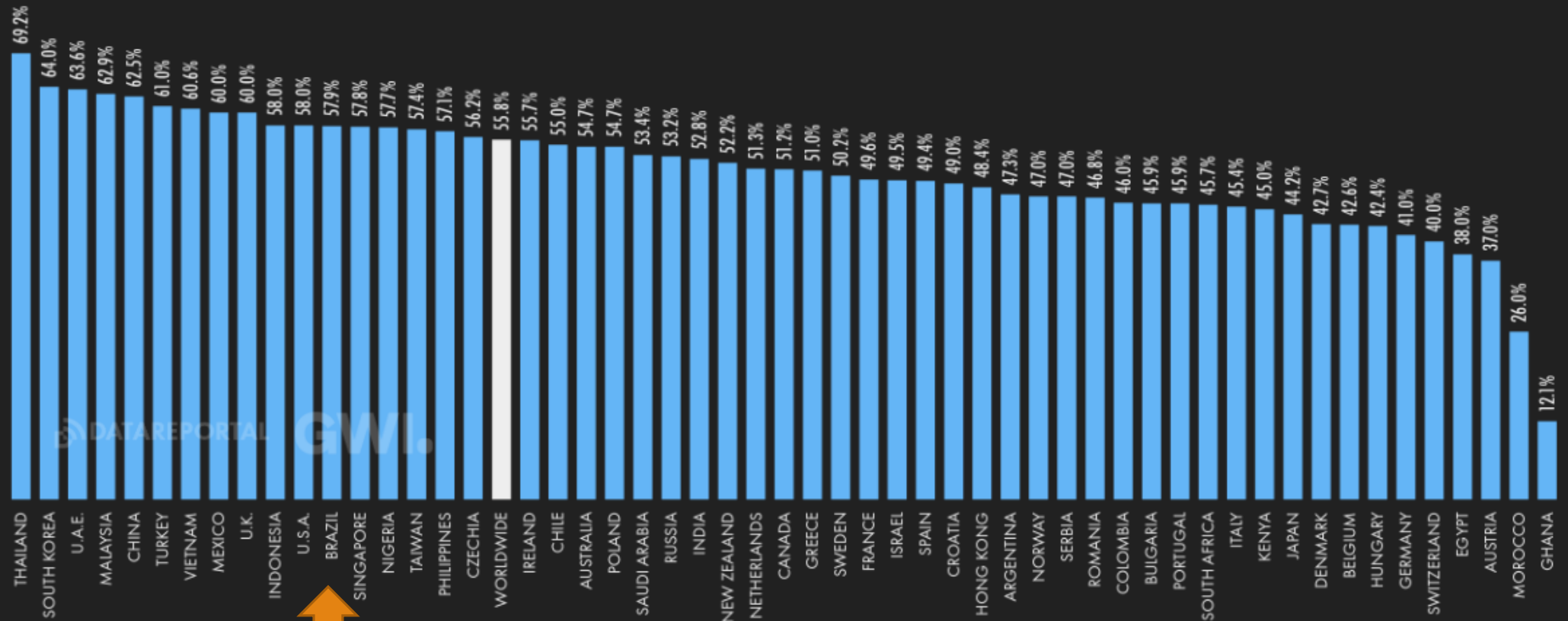
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WEEKLY ONLINE PURCHASES

PERCENTAGE OF INTERNET USERS AGED 16+ WHO BUY SOMETHING ONLINE EACH WEEK



GLOBAL OVERVIEW



Internet: Hoje

JAN
2024

ECOMMERCE: CONSUMER GOODS CATEGORIES

ESTIMATED ANNUAL SPEND IN EACH CONSUMER GOODS ECOMMERCE CATEGORY (B2C ONLY, U.S. DOLLARS, FULL-YEAR 2023)



ELECTRONICS



**\$781.3
BILLION**

YEAR-ON-YEAR CHANGE
+11.8% (+\$82 BILLION)

FASHION



**\$673.6
BILLION**

YEAR-ON-YEAR CHANGE
+8.4% (+\$52 BILLION)

FOOD



**\$370.7
BILLION**

YEAR-ON-YEAR CHANGE
+16.0% (+\$51 BILLION)

BEVERAGES



**\$209.3
BILLION**

YEAR-ON-YEAR CHANGE
+14.2% (+\$26 BILLION)

DIY & HARDWARE



**\$201.8
BILLION**

YEAR-ON-YEAR CHANGE
+8.5% (+\$16 BILLION)

FURNITURE



**\$188.0
BILLION**

YEAR-ON-YEAR CHANGE
+11.6% (+\$20 BILLION)

PHYSICAL MEDIA



**\$182.7
BILLION**

YEAR-ON-YEAR CHANGE
+0.8% (+\$1.5 BILLION)

BEAUTY &
PERSONAL CARE



**\$151.2
BILLION**

YEAR-ON-YEAR CHANGE
+5.1% (+\$7.4 BILLION)

TOBACCO
PRODUCTS



**\$104.4
BILLION**

YEAR-ON-YEAR CHANGE
+1.9% (+\$1.9 BILLION)

TOYS &
HOBBY



**\$80.53
BILLION**

YEAR-ON-YEAR CHANGE
+5.2% (+\$4.0 BILLION)

HOUSEHOLD
ESSENTIALS



**\$72.98
BILLION**

YEAR-ON-YEAR CHANGE
+11.9% (+\$7.8 BILLION)

OVER-THE-COUNTER
PHARMACEUTICALS



**\$59.65
BILLION**

YEAR-ON-YEAR CHANGE
+7.6% (+\$4.2 BILLION)

LUXURY
GOODS



**\$54.42
BILLION**

YEAR-ON-YEAR CHANGE
+9.7% (+\$4.8 BILLION)

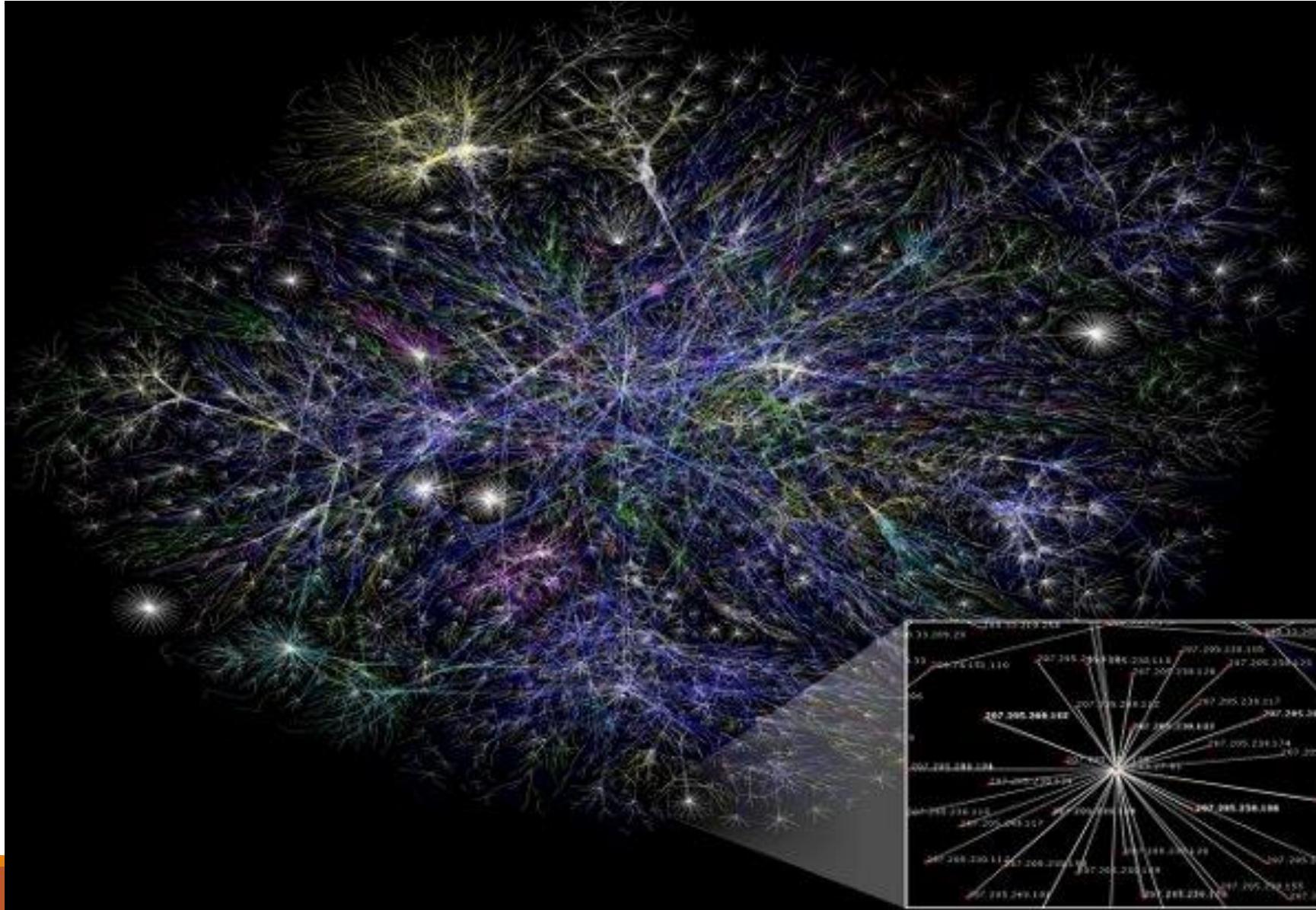
EYE-
WEAR



**\$23.48
BILLION**

YEAR-ON-YEAR CHANGE
+5.9% (+\$1.3 BILLION)

Internet: Hoje



Pontos Importantes da Videoaula

Histórico da Internet

- Conhecimentos gerais